



US012406845B2

(12) **United States Patent**
Fukui et al.

(10) **Patent No.:** **US 12,406,845 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) α -Ga₂O₃ SEMICONDUCTOR FILM

(71) Applicant: **NGK INSULATORS, LTD.**, Nagoya (JP)

(72) Inventors: **Hiroshi Fukui**, Obu (JP); **Morimichi Watanabe**, Nagoya (JP); **Jun Yoshikawa**, Nagoya (JP)

(73) Assignee: **NGK INSULATORS, LTD.**, Nagoya (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 599 days.

(21) Appl. No.: **17/653,146**

(22) Filed: **Mar. 2, 2022**

(65) **Prior Publication Data**

US 2022/0238645 A1 Jul. 28, 2022

Related U.S. Application Data

(63) Continuation of application No. PCT/JP2019/038518, filed on Sep. 30, 2019.

(51) **Int. Cl.**
H01L 21/02 (2006.01)
C23C 16/40 (2006.01)

(Continued)

(52) **U.S. Cl.**
CPC **H01L 21/02565** (2013.01); **C23C 16/40** (2013.01); **C23C 16/4481** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC H01L 21/02565; H01L 29/24; H01L 29/66969; C23C 16/40; H10D 62/405; H10D 62/80

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2015/0279944 A1* 10/2015 Hitora H01L 21/02565 257/43

2017/0179249 A1 6/2017 Oda et al.
(Continued)

FOREIGN PATENT DOCUMENTS

CN 107068773 A 8/2017
JP 2014-072533 A 4/2014

(Continued)

OTHER PUBLICATIONS

Ma, T.C. et al. "On the origin of dislocation generation and annihilation in α -Ga₂O₃ epilayers on sapphire." Appl. Phys. Lett. 115, 182101 (2019). (Year: 2019).*

(Continued)

Primary Examiner — Yara B Green

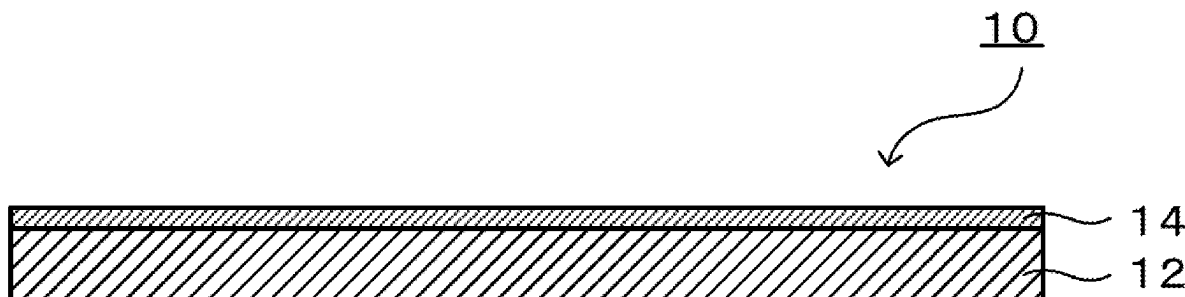
Assistant Examiner — Christopher M Roland

(74) *Attorney, Agent, or Firm* — BURR PATENT LAW, PLLC

(57) **ABSTRACT**

An α -Ga₂O₃ semiconductor film according to the present invention has a measurement point (dark spot) with a maximum emission intensity A of not more than 0.6 times the average value X of top 5% of the maximum emission intensities A at all measurement points in intensity mapping of plane cathodoluminescence, wherein the maximum emission intensity A at each measurement point is determined in the wavelength range of 250 to 365 nm.

3 Claims, 6 Drawing Sheets



(51) **Int. Cl.**

C23C 16/448 (2006.01)
H10D 62/40 (2025.01)
H10D 62/80 (2025.01)
H10D 8/60 (2025.01)
H10D 99/00 (2025.01)

FOREIGN PATENT DOCUMENTS

JP	2016-025256	A	2/2016
JP	2016-100592	A	5/2016
JP	2016-100593	A	5/2016
JP	2016-155963	A	9/2016
JP	2017-118090	A	6/2017
JP	2019-033142	A	2/2019

(52) **U.S. Cl.**

CPC **H01L 21/02414** (2013.01); **H01L 21/0242**
 (2013.01); **H01L 21/0243** (2013.01); **H01L**
21/0262 (2013.01); **H01L 21/02628** (2013.01);
H10D 62/405 (2025.01); **H10D 62/80**
 (2025.01); **H10D 8/60** (2025.01); **H10D 99/00**
 (2025.01)

OTHER PUBLICATIONS

Stepanov, S.I. et al. "HVPE growth of corundum-structured α -Ga₂O₃ on sapphire substrates with α -Cr₂O₃ buffer layer." *Materials Physics and Mechanics* 47 (2021) 577-581. (Year: 2021).*

Polyakov, A. et al. "Effects of sapphire substrate orientation on Sn-doped α -Ga₂O₃ growth by halide vapor phase epitaxy using α -Ga₂O₃ buffers." *J. Phys. D: Appl. Phys.* 55 (2022) 495102 (6pp). (Year: 2022).*

Dang, G.T. et al. "Growth of α -Cr₂O₃ single crystals by mist CVD using ammonium dichromate." *Applied Physics Express* 11, 111101 (2018). (Year: 2018).*

English translation of the International Preliminary Report on Patentability (Chapter I) dated Apr. 14, 2022 (Application No. PCT/JP2019/038518).

International Search Report and Written Opinion (Application No. PCT/JP2019/038518) dated Dec. 3, 2019.

Chinese Office Action dated Jun. 29, 2023 (Application No. 201980098097.5).

(56)

References Cited

U.S. PATENT DOCUMENTS

2018/0061952	A1 *	3/2018	Tokuda		H01L 21/0242
2019/0057866	A1 *	2/2019	Oshima		C30B 29/16
2019/0067426	A1 *	2/2019	Sugimoto		H01L 21/02576
2021/0226002	A1 *	7/2021	Takahashi		H01L 21/0262
2021/0404089	A1 *	12/2021	Watanabe		H01L 21/02658
2021/0404090	A1 *	12/2021	Watanabe		H01L 21/0254
2021/0408242	A1 *	12/2021	Watanabe		C30B 29/16
2022/0029022	A1 *	1/2022	Watanabe		C23C 16/029
2022/0059424	A1 *	2/2022	Hashigami		H01L 21/02565
2024/0003043	A1 *	1/2024	Yoshikawa		H10D 8/00

* cited by examiner

Fig. 1A

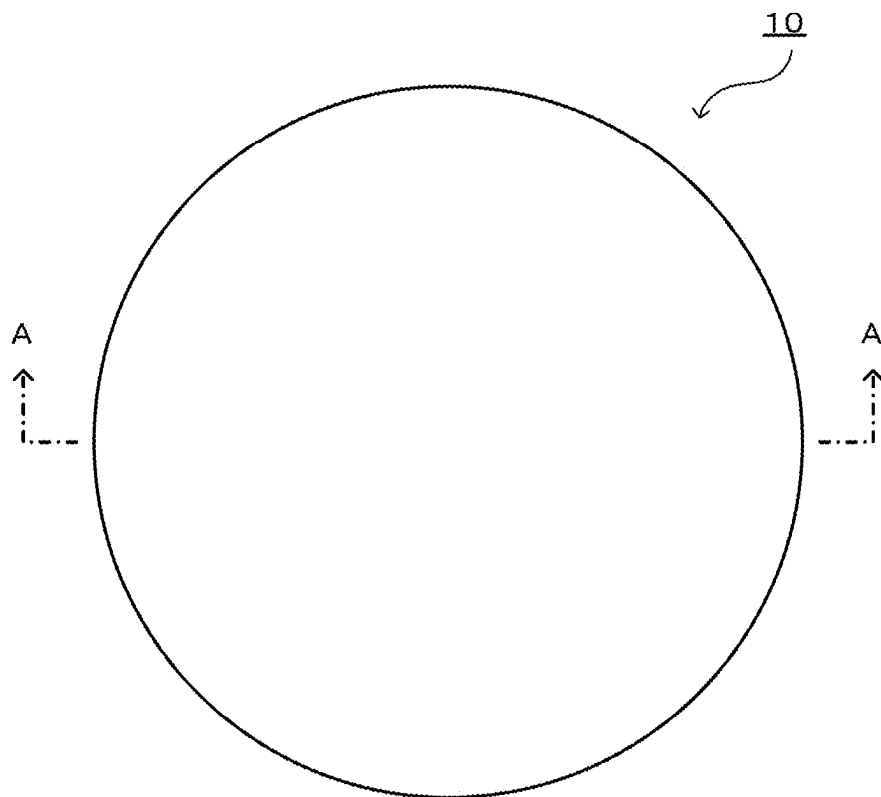


Fig. 1B

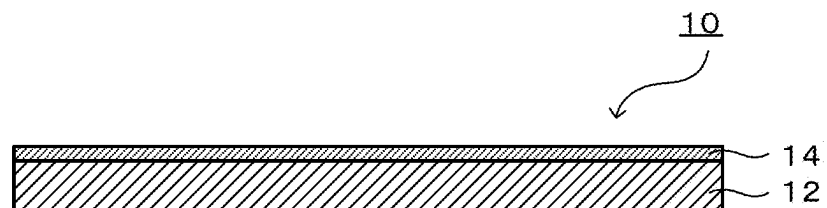


Fig. 2

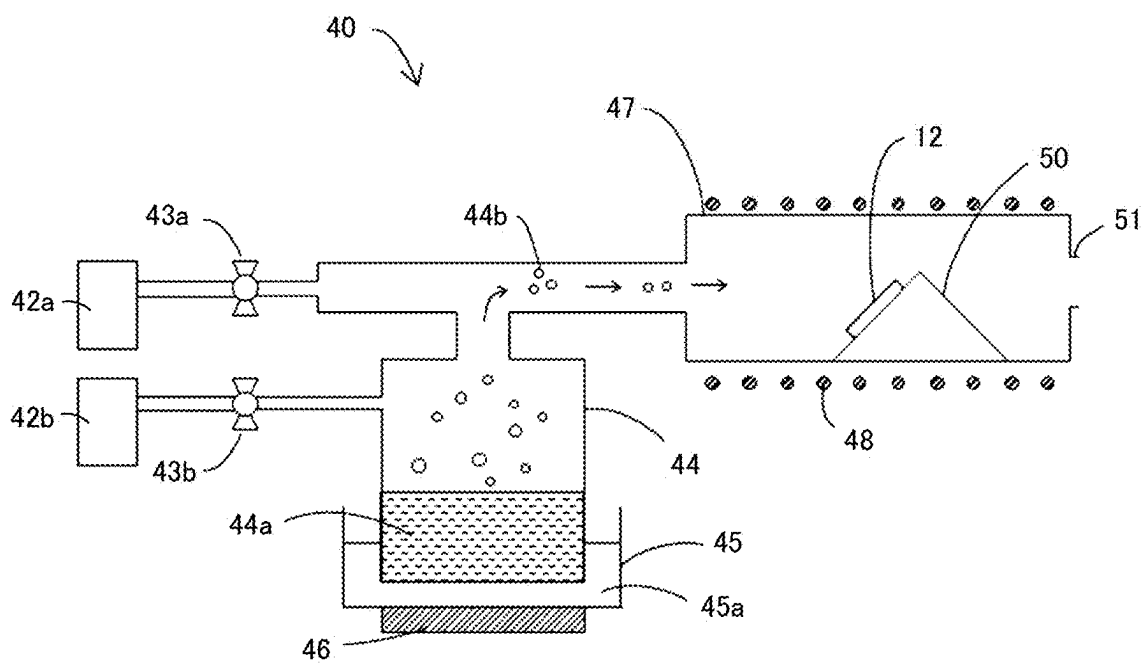


Fig. 3

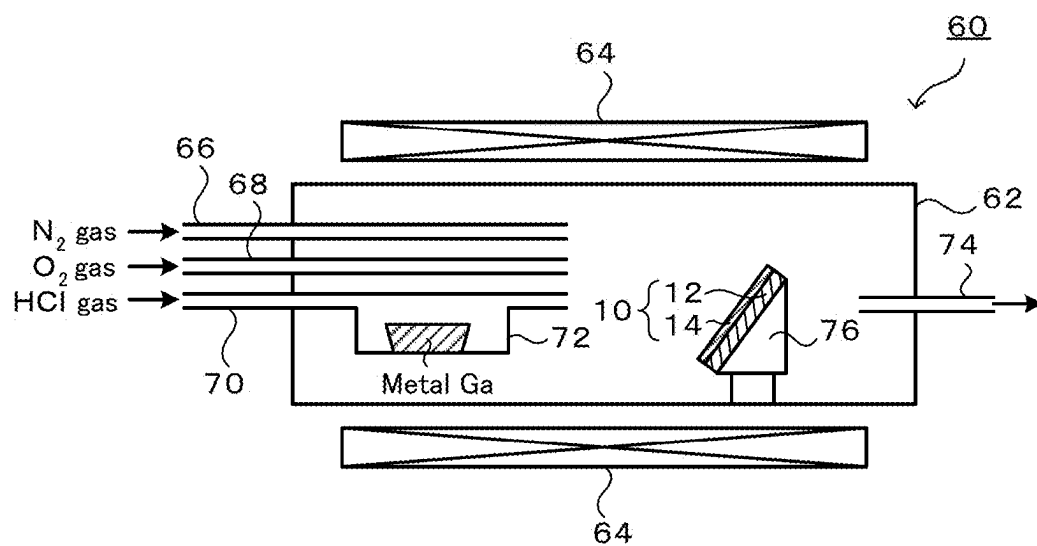


Fig. 4

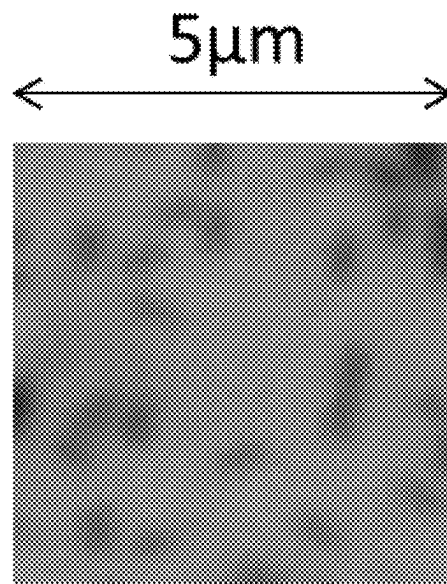


Fig. 5

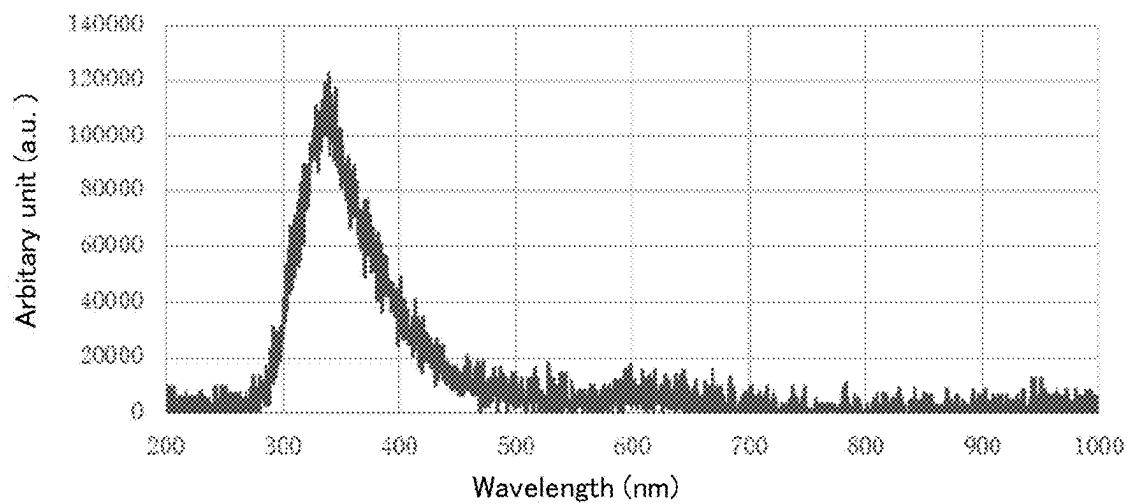


Fig. 6

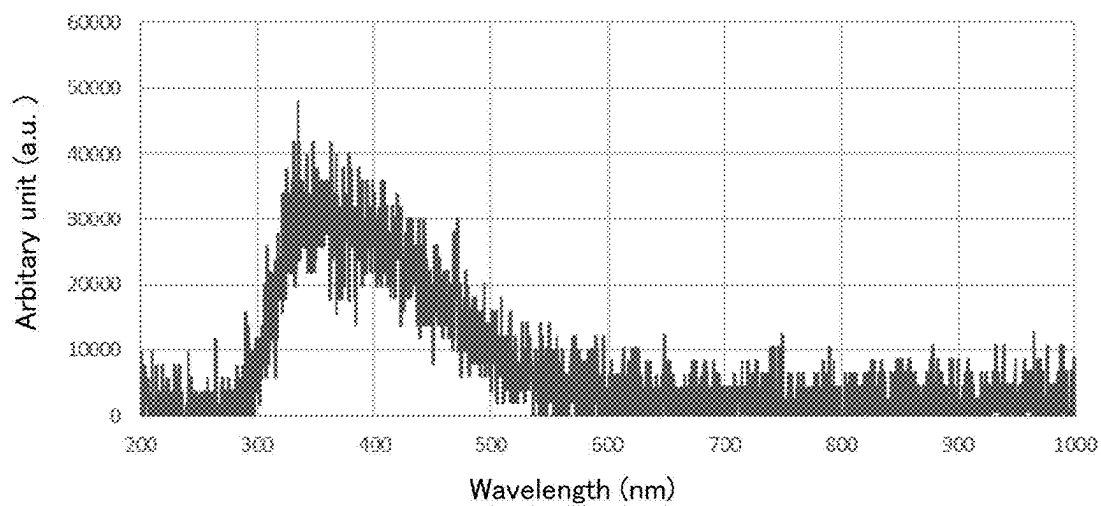
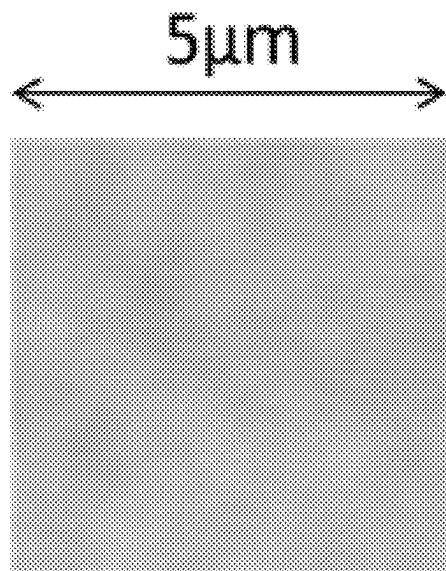


Fig. 7



1

 α -Ga₂O₃ SEMICONDUCTOR FILM

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to an α -Ga₂O₃ semiconductor film.

2. Description of the Related Art

In recent years, gallium oxide (Ga₂O₃) has attracted attention as a semiconductor material. Gallium oxide is known to have five crystal forms of α , β , γ , δ , and E, among which α -Ga₂O₃ has a very large band gap of 5.3 eV and is expected as a power semiconductor material. Because α -Ga₂O₃ is a metastable phase, however, α -Ga₂O₃ has not been practically used as a single-crystal substrate and is generally formed on a sapphire substrate by heteroepitaxial growth.

For example, PTL 1 discloses a semiconductor device including a base substrate with a corundum crystal structure, a semiconductor layer with a corundum crystal structure, and an insulating film with a corundum crystal structure, and describes an example in which an α -Ga₂O₃ film is formed as a semiconductor layer on a sapphire substrate. PTL 2 discloses a semiconductor device including an n-type semiconductor layer composed mainly of a crystalline oxide semiconductor with a corundum structure, a p-type semiconductor layer composed mainly of an inorganic compound with a hexagonal crystal structure, and an electrode. PTL 2 discloses in an example that a diode is produced by forming an α -Ga₂O₃ film with a corundum structure, which is a metastable phase, as an n-type semiconductor layer and an α -Rh₂O₃ film with a hexagonal crystal structure as a p-type semiconductor layer on a C-plane sapphire substrate.

However, there is a problem that a crack or a crystal defect is formed when an α -Ga₂O₃ film is grown on different substrates. When an InAlGaO semiconductor film, which is a mixed crystal of α -Ga₂O₃ and a different corundum material, is formed, in general, there is a cracking problem in an epitaxial film due to crystal growth on different substrates. As a technique to solve this problem, PTL 3 discloses the production of an α -Ga₂O₃ film with fewer cracks. PTL 4 discloses that an α -Ga₂O₃ film with fewer cracks is produced by forming voids during the formation of an epitaxial film. PTL 5 shows a result of cross-sectional CL intensity mapping of an α -Ga₂O₃ film using a cathodoluminescence (CL) spectrometer (a graph showing the relationship between the wavelength and emission intensity).

CITATION LIST

Patent Literature

PTL 1: JP 2014-072533 A
 PTL 2: JP 2016-025256 A
 PTL 3: JP 2016-100592 A
 PTL 4: JP 2016-100593 A
 PTL 5: JP 2016-155963 A

SUMMARY OF THE INVENTION

Although various α -Ga₂O₃ films have been produced as described above, it has been desired to develop a useful α -Ga₂O₃ semiconductor film different from known films.

2

The present invention has been made to solve such problems and mainly aims to provide a novel α -Ga₂O₃ semiconductor film.

An α -Ga₂O₃ semiconductor film according to the present invention has a measurement point with a maximum emission intensity A of not more than 0.6 times the average value X of top 5% of the maximum emission intensities A at all measurement points in intensity mapping of plane CL, wherein the maximum emission intensity A at each measurement point is determined in the wavelength range of 250 to 365 nm. A measurement point with a maximum emission intensity A of not more than 0.6 times the X is hereinafter referred to as a dark spot.

This α -Ga₂O₃ semiconductor film is a novel semiconductor film and has fewer cracks.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A and 1B are explanatory views of a multilayer structure 10, wherein FIG. 1A is a plan view, and FIG. 1B is a cross-sectional view taken along the line A-A.

FIG. 2 is a schematic cross-sectional view of the structure of a mist CVD apparatus 40.

FIG. 3 is a schematic cross-sectional view of the structure of a vapor-phase growth apparatus 60.

FIG. 4 is a plane CL intensity mapping image of Example 1.

FIG. 5 is a graph of an example of a CL spectrum at a measurement point within the top 5% of the maximum emission intensities A at all measurement points in Example 1.

FIG. 6 is a graph of an example of a CL spectrum at a dark spot in Example 1.

FIG. 7 is a plane CL intensity mapping image of Comparative Example 1.

DETAILED DESCRIPTION OF THE INVENTION

[Multilayer Structure]

FIGS. 1A and 1B are explanatory views of a multilayer structure 10, wherein 1A is a plan view, and 1B is a cross-sectional view taken along the line A-A.

The multilayer structure 10 is a plate-like member and has a semiconductor film 14 on a base substrate 12. When viewed in a plan view, the multilayer structure 10 in the present embodiment is circular. The "circular" is not necessarily perfect circular and may also be generally circular that can be generally recognized as circular as a whole. For example, the circle may be partly cut out to specify the crystal orientation or for another purpose. When viewed in a plan view, the multilayer structure 10 is not limited to circular and may be polygonal (tetragonal, such as square or rectangular, pentagonal, hexagonal, etc.).

The base substrate 12 preferably has a layer of an oxide (α -Cr₂O₃, α -Fe₂O₃, etc.) with a lattice constant closer to α -Ga₂O₃ than sapphire and more preferably has a single-crystal layer of α -Cr₂O₃ or α -Cr₂O₃ solid solution.

The semiconductor film 14 is a semiconductor film with a corundum crystal structure formed of α -Ga₂O₃ or α -Ga₂O₃ solid solution, that is, an α -Ga₂O₃ semiconductor film. α -Ga₂O₃ belongs to the trigonal crystal group and has a corundum crystal structure. α -Ga₂O₃ solid solution is a solid solution of another component dissolved in α -Ga₂O₃ and maintains the corundum crystal structure. The other component is Al₂O₃, In₂O₃, Cr₂O₃, Fe₂O₃, Rh₂O₃, V₂O₃, or Ti₂O₃, for example.

The semiconductor film **14** includes a dark spot, which is defined as a measurement point with a maximum emission intensity *A* of not more than 0.6 times the average value *X* of the top 5% of the maximum emission intensities *A* at all measurement points in intensity mapping of plane CL, wherein the maximum emission intensity *A* at each measurement point is determined in the wavelength range of 250 to 365 nm. The semiconductor film **14** has fewer cracks. The physical meaning of a dark portion of a plane CL mapping image detected as a dark spot in an α -Ga₂O₃ semiconductor film has not been clarified, and the reason for the effect of suppressing cracking is also unknown. For example, an α -Ga₂O₃ film formed directly on sapphire has a large number of edge dislocations and screw dislocations but has no dark spot detected in a CL mapping image in the wavelength range of 250 to 365 nm. A dark spot defined in the present application is therefore not caused by these dislocations. Thus, crystal defects not attributable to edge dislocations, screw dislocations, mixed dislocations, or the like may be present, and such defects may cause dark spots. Such a defect may relieve the stress caused by film formation and suppress cracking in a semiconductor film.

Furthermore, the ratio *R* of the number of dark spots to the number of all measurement points in the semiconductor film **14** is preferably smaller, preferably 0.600 or less, more preferably 0.200 or less, in terms of electrical characteristics, for example, to reduce leakage current when the semiconductor film **14** is used for a Schottky barrier diode. Although a dark spot is considered to be a defect not attributable to a dislocation, as described above, an excessive number of dark spots affect electrical characteristics. On the other hand, to suppress cracking in a semiconductor film, *R* is preferably larger, preferably 0.005 or more, more preferably 0.050 or more. Thus, to reduce leakage current and cracking in a Schottky barrier diode, *R* preferably ranges from 0.005 to 0.600, more preferably 0.050 to 0.200.

The measurement conditions for intensity mapping of plane CL are described below. Emission intensity in the intensity mapping measurement is expressed in arbitrary unit (a.u.).

Measurement range: 5 μ m $\times$ 5 μ m of a plane (surface) of the semiconductor film **14**, 20 fields

Accelerating voltage: 10 kV

Measurement temperature: room temperature

Diffraction grating of spectrometer: 100 gr/mm, blaze wavelength 450 nm

Number of CL image pixels: 51 $\times$ 51=2601

(A 5 μ m $\times$ 5 μ m square area was divided into 2601 regions to measure CL intensity. One measurement point is calculated to be an approximately 98 nm $\times$ approximately 98 nm region.)

The area of a surface of the semiconductor film **14** is substantially the same as the area of the base substrate **12**. The area of a surface of the semiconductor film **14** is preferably 20 cm² or more, more preferably 70 cm² or more, still more preferably 170 cm² or more. The area of the semiconductor film **14** can be increased in this manner to produce a large number of semiconductor elements from one semiconductor film **14** and thereby reduce production costs. The upper limit of the size of the semiconductor film **14** is typically, but not limited to, 700 cm² or less on one side. The semiconductor film **14** preferably has an average thickness of 3 μ m or more, more preferably 5 μ m or more, still more preferably 8 μ m or more.

The semiconductor film **14** can contain 1.0×10^{16} to 1.0×10^{21} /cm³ of a group 14 element as a dopant. The group 14 element herein refers to a group 14 element in the periodic

table of the International Union of Pure and Applied Chemistry (IUPAC), more specifically, any one element of carbon (C), silicon (Si), germanium (Ge), tin (Sn), and lead (Pb). The amount of the dopant may depend on the desired characteristics and preferably ranges from 1.0×10^{16} to 1.0×10^{21} /cm³, more preferably 1.0×10^{17} to 1.0×10^{19} /cm³. Preferably, the dopant is uniformly distributed in the film, and the dopant concentration of the semiconductor film **14** is almost the same on the front and back surfaces.

Furthermore, the semiconductor film **14** is preferably an alignment film oriented in a specific plane direction. The orientation of the semiconductor film **14** can be examined by a known method, for example, by inverse pole figure orientation mapping with an electron beam backscatter diffraction device (EBSD). For example, the semiconductor film may be oriented in the C-axis direction or may be oriented in the C-axis direction and in the in-plane direction.

[Method for Producing Semiconductor Film]

A method for producing the semiconductor film **14** includes (a) the step of forming the multilayer structure **10** and (b) the step of separating the semiconductor film **14** from the base substrate **12**.

(a) Step of Forming Multilayer Structure **10**

The multilayer structure **10** is produced by forming the semiconductor film **14** on the base substrate **12** formed of an oxide (α -Cr₂O₃, α -Fe₂O₃, etc.) with a lattice constant closer to α -Ga₂O₃ than sapphire. The base substrate **12** is a Cr₂O₃ single-crystal substrate, for example. The film-forming method may be, but is not limited to, a known method. The film-forming method is preferably mist CVD, HVPE, MBE, MOCVD, sputtering, or a hydrothermal method, more preferably mist CVD or HVPE, still more preferably mist CVD.

FIG. 2 is a schematic cross-sectional view of the structure of a mist CVD apparatus **40**. The mist CVD apparatus **40** includes a susceptor **50** on which the base substrate **12** is to be placed, a diluent gas source **42a**, a carrier gas source **42b**, a flow control valve **43a** for adjusting the flow rate of diluent gas from the diluent gas source **42a**, a flow control valve **43b** for adjusting the flow rate of carrier gas from the carrier gas source **42b**, a mist generation source **44** for containing a raw material solution **44a**, a container **45** for water **45a**, an ultrasonic transducer **46** on the bottom of the container **45**, a quartz tube **47** serving as a film formation chamber, a heater **48** around the quartz tube **47**, and an exhaust port **51**. The susceptor **50** is made of quartz and has an inclined surface on which the base substrate **12** is to be placed.

The raw material solution **44a** for use in the mist CVD method may be any solution for an α -Ga₂O₃ semiconductor film, for example, a solution of at least one of organometallic complexes of Ga, Ga halides, and organometallic complexes of Ga and a metal forming a solid solution, dissolved in a solvent. Examples of the organometallic complexes include acetylacetonate complexes, and examples of the halides include GaCl₃ and GaBr₃. When a halide is dissolved in a solvent as a raw material, the raw material solution contains a halogen element. To facilitate the preparation for the solution, the halide may be dissolved directly in the solvent or may be dissolved in water to which a hydrohalic acid (for example, hydrochloric acid) is added, or metal Ga may be dissolved in a solution of water and a hydrohalic acid. To form an α -Ga₂O₃ semiconductor film containing a group 14 element as a dopant or to form a mixed crystal film with α -Ga₂O₃ containing an oxide of In or Al, a solution containing these components may be added to the raw material solution. Furthermore, when an organometallic complex is used as a raw material, an additive agent, such as a hydro-

5

halic acid, may be added to the raw material solution. The solvent may be water or an alcohol, etc.

Next, the raw material solution **44a** is atomized to generate a mist **44b**. A preferred example of the atomization method is a method of vibrating the raw material solution **44a** with the ultrasonic transducer **46**. The mist **44b** is then transported to the film formation chamber using a carrier gas. The carrier gas may be, but is not limited to, one or two or more of inert gases, such as oxygen, ozone, and nitrogen, and reducing gases, such as hydrogen.

The base substrate **12** is placed in the film formation chamber (the quartz tube **47**). The mist **44b** transported to the film formation chamber is subjected to thermal decomposition and a chemical reaction and forms a film on the base substrate **12**. Depending on the type of raw material solution, the reaction temperature preferably ranges from 300° C. to 800° C., more preferably 350° C. to 700° C. The atmosphere in the film formation chamber is not particularly limited as long as a desired semiconductor film is formed, may be an oxygen gas atmosphere, an inert gas atmosphere, a vacuum, or a reducing atmosphere, and is preferably an air atmosphere. Instead of or in addition to the mist **44b**, a droplet of the raw material solution **44a** may be used.

FIG. 3 is a schematic cross-sectional view of the structure of a vapor-phase growth apparatus **60** using HVPE. The vapor-phase growth apparatus **60** includes a reaction vessel **62** and a heater **64**.

The reaction vessel **62** is made of a material that does not react with raw materials or products (for example, quartz). A carrier gas supply pipe **66**, an oxidizing gas supply pipe **68**, and a raw material supply pipe **70** are attached to one of opposite side surfaces of the reaction vessel **62**, and an exhaust pipe **74** is attached to the other side surface. A carrier gas (for example, nitrogen, a noble gas, etc.) is supplied to the reaction vessel **62** through the carrier gas supply pipe **66**. Oxygen gas is supplied as an oxidizing gas to the reaction vessel **62** through the oxidizing gas supply pipe **68**. The oxidizing gas other than oxygen may be water vapor or dinitrogen monoxide. In the raw material supply pipe **70**, a halogen gas (for example, chlorine gas) or a hydrogen halide gas (for example, hydrogen chloride gas) supplied from a gas supply source reacts with metal gallium in a storage portion **72** provided in the middle of the raw material supply pipe **70** and generates gallium halide. Thus, the gallium halide gas is supplied as a raw material gas to the reaction vessel **62** through the raw material supply pipe **70**. The halogen gas or hydrogen halide gas may be supplied together with a carrier gas, such as nitrogen or a noble gas. A susceptor **76** for detachably holding the base substrate **12** is located downstream of the supply pipes **66**, **68**, and **70** in the reaction vessel **62**. Unreacted gases are discharged from the reaction vessel **62** through the exhaust pipe **74**. The exhaust pipe **74** may be coupled to a vacuum pump, which adjust the degree of vacuum in the reaction vessel **62**. This can suppress a gas-phase reaction or improve the growth rate distribution.

The heater **64** surrounds the reaction vessel **62**. A resistance heater can be adopted as the heater **64**, for example.

Production of the multilayer structure **10** according to the present embodiment using the vapor-phase growth apparatus **60** is described below. In the reaction vessel **62**, the oxygen gas from the oxidizing gas supply pipe **68** reacts with the raw material gas (gallium halide gas) from the raw material supply pipe **70** and forms the semiconductor film **14** (α -Ga₂O₃ film) on the base substrate **12**. The film-forming temperature is not particularly limited and may be appropriately set in the range of 300° C. to 800° C. The partial

6

pressure of the oxygen gas or the raw material gas is not particularly limited. For example, the partial pressure of the raw material gas may range from 0.05 to 10 kPa, and the partial pressure of the oxygen gas may range from 0.25 to 50 kPa. The growth time may be appropriately determined according to the designed thickness of the semiconductor film **14**. The multilayer structure **10** is thus formed.

(b) Step of Separating Semiconductor Film **14** from Base Substrate **12**

The semiconductor film **14** can be separated from the base substrate **12** of the multilayer structure **10** at room temperature by any method, including a known method. The separation method may be a separation method with mechanical impact, a separation method of applying heat to utilize thermal stress, or a separation method of applying vibration, such as ultrasonic waves. The semiconductor film **14** can be separated as a free-standing film. Alternatively, the semiconductor film **14** may be transferred onto another supporting substrate.

The above-described α -Ga₂O₃ semiconductor film according to the present embodiment is a hitherto unknown novel film and can improve the device characteristics due to its high dielectric breakdown electric field characteristics. For example, the α -Ga₂O₃ semiconductor film can be used for a Schottky barrier diode to reduce leakage current.

It goes without saying that the present invention should not be limited to these embodiments and can be implemented in various aspects within the technical scope of the present invention.

EXAMPLES

Examples according to the present invention are described below. It should be noted that the present invention is not limited to these examples.

Example 1

The mist CVD apparatus **40** illustrated in FIG. 2 was used to form an α -Ga₂O₃ film (semiconductor film) by the following method. The base substrate **12** was a commercial Cr₂O₃ single crystal (8 mm×8 mm, 0.5 mm in thickness, C-plane, no off-angle, hereinafter referred to as a Cr₂O₃ substrate).

(1) Formation of α -Ga₂O₃ Film by Mist CVD Method (1a) Adjustment of Raw Material Solution

An aqueous solution was adjusted at a gallium acetylacetonate concentration of 0.08 mol/L. The aqueous solution contained 1.5% by volume of 36% hydrochloric acid. Tin(II) chloride dihydrate (SnCl₂·2H₂O) was added to the aqueous solution such that the atomic ratio of tin to gallium was 0.2. The aqueous solution was used as the raw material solution **44a**.

(1b) Preparation of Film Formation

Next, the raw material solution **44a** was poured into the mist generation source **44** of the mist CVD apparatus **40** illustrated in FIG. 2. A Cr₂O₃ substrate was placed on the susceptor **50** as the base substrate **12**, and the temperature in the quartz tube **47** was increased to 580° C. with the heater **48**. The flow control valves **43a** and **43b** were then opened to supply a diluent gas and a carrier gas to the quartz tube **47** from the diluent gas source **42a** and the carrier gas source **42b**. After the atmosphere in the quartz tube **47** was sufficiently replaced with the diluent gas and the carrier gas, the flow rate of the diluent gas was adjusted to 0.5 L/min, and the flow rate of the carrier gas was adjusted to 1 L/min. Nitrogen gas was used as the diluent gas and the carrier gas.

(1c) Film Formation

Next, the ultrasonic transducer **46** was vibrated at 2.4 MHz, and the vibration was transmitted through the water **45a** to the raw material solution **44a** to form the mist **44b** of the raw material solution **44a**. The mist **44b** was introduced with the diluent gas and the carrier gas into the quartz tube **47** serving as a film formation chamber, was reacted in the quartz tube **47**, and formed a crystalline α -Ga₂O₃ semiconductor film (semiconductor layer) on the base substrate **12** by a CVD reaction on the surface of the base substrate **12**. The film-forming time was 40 minutes. Thus, the multilayer structure **10** with the semiconductor film **14** on the base substrate **12** was formed.

(2) Evaluation of Semiconductor Film

(2a) Surface EDS

EDS measurement of a film surface on the film formation side of the resulting film showed that only Ga and O were detected, and the film was Ga oxide.

(2b) EBSD

A SEM (SU-5000, manufactured by Hitachi High-Technologies Corporation) equipped with an electron beam backscatter diffraction apparatus (EBSD) (Nordlys Nano, manufactured by Oxford Instruments) was used to perform inverse pole figure orientation mapping of the film surface on the film formation side composed of Ga oxide in a 500 μm ×500 μm field. The conditions for the EBSD measurement were described below.

<EBSD Measurement Conditions>

Accelerating voltage: 15 kV

Spot intensity: 70

Working distance: 22.5 mm

Step size: 0.5 μm

Sample tilt angle: 70 degrees

Measurement program: Aztec (version 3.3)

The inverse pole figure orientation mapping showed that the Ga oxide film had a biaxially-oriented corundum crystal structure with C-axis orientation in the direction normal to the substrate and with in-plane orientation. This showed the formation of an alignment film formed of α -Ga₂O₃.

(2c) Measurement of Plane CL Intensity Mapping

The film was subjected to a plane CL intensity mapping measurement using a CL spectrometer (spectrometer: iHR-320 manufactured by Horiba, Ltd., SEM: Schottky emission-type SEM JSM-7100F/TTLS manufactured by JEOL Ltd.) in twenty fields with a 5 μm ×5 μm field. The plane CL intensity mapping measurement was performed under the following conditions.

<Plane CL Intensity Mapping Measurement Conditions>

Detector: CCD (Jobin Yvon)

Diffraction grating of spectrometer: 100 gr/mm, blaze wavelength 450 nm

Number of CL image pixels: 51×51

CL spectrum integration time: 10 ms×1

Measurement temperature: Room temperature

Irradiation current: 4.8 nA

Accelerating voltage: 10 kV

W.D.: 10.3 mm

Number of SEM image pixels: 250×250

The value (ratio R) of (the number of dark spots)/(the number of all measurement points) calculated from all CL image measurement points was 0.053. FIG. 4 shows a mapping image obtained by integrating emission intensities in the wavelength range of 277 to 361 nm in a plane CL intensity mapping measurement. The maximum emission intensity A in the wavelength range of 250 to 365 nm was determined at each measurement point in the plane CL intensity mapping measurement, and the average value X of

the top 5% of the maximum emission intensities A at all measurement points was determined. A measurement point with a maximum emission intensity A of not more than 0.6 times the X was defined as a dark spot. FIG. 5 shows an example of a CL spectrum at a measurement point within the top 5% of the maximum emission intensities A at all measurement points. FIG. 6 shows an example of a CL spectrum at a dark spot

(2d) Appearance Evaluation of Semiconductor Film

The appearance of the semiconductor film was observed in a polarization/differential interference mode over the entire surface with an industrial microscope (ECLIPSE LV150N manufactured by Nikon Corporation) equipped with a 10× ocular lens and a 5× objective lens, and no cracks were observed.

(3) Production and Evaluation of Device

After a 2 mm×8 mm region in an end portion of the semiconductor film **14** (8 mm×8 mm) of the multilayer structure **10** formed in (1c) was masked with a sapphire substrate (not shown), an n- layer was formed. The n- layer was formed in the same manner as in (1) except that a dopant (more specifically, tin(II) chloride) was not added in the preparation of the raw material solution in (1a) and the film-forming time was 20 minutes. After the n- layer was formed, the masking sapphire substrate (not shown) was removed to expose an n+ layer. A Ti electrode (ohmic electrode) was formed in the exposed region of the n+ layer, and a Pt electrode (Schottky electrode) was formed on the n-layer. Thus, a lateral Schottky barrier diode was produced. Application of a predetermined voltage to the Schottky barrier diode in the reverse direction caused a negligible leakage current, thus showing good device characteristics.

Although the Schottky barrier diode in the present example was a lateral device to simply evaluate the device characteristics, a vertical device produced by removing the base substrate **12** also has the same tendency as the lateral device according to the present example.

Example 2

An α -Ga₂O₃ film was formed in the same manner as in Example 1 except that the temperature in the quartz tube **47** at which the α -Ga₂O₃ film was formed by the mist CVD method was 430° C. Various evaluations were performed. The ratio R was 0.192. The entire film surface was observed in the same manner as in Example 1, and no cracks were observed. Application of a predetermined voltage to the Schottky barrier diode produced in the same manner as in Example 1 in the reverse direction caused a negligible leakage current, thus showing good device characteristics, as in Example 1.

Example 3

An α -Ga₂O₃ film was formed in the same manner as in Example 1 except that a surface of a commercial Cr₂O₃ single crystal (8 mm×8 mm, 0.5 mm in thickness, C-plane, no off-angle) serving as a base substrate for film formation was ground to have a surface roughness Ra of 50 nm and the temperature in the quartz tube **47** at which the α -Ga₂O₃ film was formed by the mist CVD method was 430° C. Various evaluations were performed. The ratio R was 0.583. The entire film surface was observed in the same manner as in Example 1, and no cracks were observed. Application of a predetermined voltage to the Schottky barrier diode pro-

duced in the same manner as in Example 1 in the reverse direction caused a leakage current slightly larger than that in Example 1.

Example 4

An α -Ga₂O₃ film was formed in the same manner as in Example 1 except that a surface of a commercial Cr₂O₃ single crystal (8 mm×8 mm, 0.5 mm in thickness, C-plane, no off-angle) serving as a base substrate for film formation was ground to have a surface roughness Ra of 170 nm and the temperature in the quartz tube 47 at which the α -Ga₂O₃ film was formed by the mist CVD method was 390° C. Various evaluations were performed. The ratio R was 0.692. The entire film surface was observed in the same manner as in Example 1, and no cracks were observed. Application of a predetermined voltage to the Schottky barrier diode produced in the same manner as in Example 1 in the reverse direction caused a leakage current larger than that in Example 3.

Example 5

An α -Ga₂O₃ film was formed in the same manner as in Example 1 except that the gallium acetylacetonate concentration was 0.07 mol/L, the flow rate of the diluent gas was 0.4 L/min, the flow rate of the carrier gas was 0.8 L/min, the temperature in the quartz tube 47 was 615° C., and the film-forming time was 60 minutes in the formation of the α -Ga₂O₃ film by the mist CVD method. Various evaluations were performed. The ratio R was 0.008. The entire film surface was observed in the same manner as in Example 1, and a small number of cracks were observed. Application of a predetermined voltage to the Schottky barrier diode produced in the same manner as in Example 1 in the reverse direction caused a negligible leakage current, thus showing good device characteristics, as in Example 1.

Comparative Example 1

An α -Ga₂O₃ film was formed in the same manner as in Example 1 except that a C-plane sapphire substrate 5.08 cm (2 inch) in diameter and 0.65 mm in thickness was used as a base substrate for film formation and the temperature in the quartz tube 47 at which the α -Ga₂O₃ film was formed by the mist CVD method was 430° C. Various evaluations were performed. FIG. 7 shows a mapping image obtained by integrating emission intensities in the wavelength range of 277 to 361 nm in a plane CL intensity mapping measurement. The ratio R was 0.000. In FIG. 7, the contrast of the mapping image is adjusted to emphasize the emission intensity difference within the CL measurement range. Although FIG. 7 therefore appears brighter than FIG. 4, the actual emission intensity is smaller than that in Example 1. The entire film surface was observed in the same manner as in Example 1, and a large number of cracks were observed. Application of a predetermined voltage to the Schottky barrier diode produced in the same manner as in Example 1 in the reverse direction caused a leakage current much larger than that in Example 4.

Comparative Example 2

(1) Formation of α -Cr₂O₃ Film

The mist CVD apparatus 40 illustrated in FIG. 2 was used to form an α -Cr₂O₃ film on the surface of a sapphire

substrate (50.8 mm (2 inches) in diameter, 0.43 mm in thickness, C-plane, off-angle of 0.3 degrees) by the following method.

(1a) Adjustment of Raw Material Solution

An aqueous solution was adjusted at an ammonium dichromate concentration of 0.1 mol/L and was used as the raw material solution 44a.

(1b) Preparation of Film Formation

Next, the raw material solution 44a was poured into the mist generation source 44 of the mist CVD apparatus 40 illustrated in FIG. 2. A sapphire substrate was placed on the susceptor 50 as the base substrate 12, and the temperature in the quartz tube 47 was increased to 420° C. with the heater 48. The flow control valves 43a and 43b were then opened to supply a diluent gas and a carrier gas to the quartz tube 47 from the diluent gas source 42a and the carrier gas source 42b. After the atmosphere in the quartz tube 47 was sufficiently replaced with the diluent gas and the carrier gas, the flow rate of the diluent gas was adjusted to 2.2 L/min, and the flow rate of the carrier gas was adjusted to 4.8 L/min. Nitrogen gas was used as the diluent gas and the carrier gas.

(1c) Film Formation

Next, the ultrasonic transducer 46 was vibrated at 2.4 MHz, and the vibration was transmitted through the water 45a to the raw material solution 44a to form the mist 44b of the raw material solution 44a. The mist 44b was introduced with the diluent gas and the carrier gas into the quartz tube 47 serving as a film formation chamber, was reacted in the quartz tube 47, and formed an oxide deposited layer on the base substrate 12 by a CVD reaction on the surface of the base substrate 12. The film-forming time was 50 minutes. Thus, a composite base substrate with the oxide deposited layer on the sapphire substrate 12 was formed.

(1d) Surface EDX

An energy dispersive X-ray analyzer (EDX) was used to analyze the composition of the composite base substrate surface. The results showed that only Cr and O were detected, and the oxide deposited layer was Cr oxide.

(1e) Surface EBSD

A SEM (SU-5000, manufactured by Hitachi High-Technologies Corporation) equipped with an electron beam backscatter diffraction apparatus (EBSD) (Nordlys Nano, manufactured by Oxford Instruments) was used to perform inverse pole figure orientation mapping of the substrate surface composed of the Cr oxide layer in a 500 μ m×500 μ m field. The conditions for the EBSD measurement were described below.

<EBSD Measurement Conditions>

Accelerating voltage: 15 kV

Spot Intensity: 70

Working distance: 22.5 mm

Step size: 0.5 μ m

Sample tilt angle: 70 degrees

Measurement program: Aztec (version 3.3)

The inverse pole figure orientation mapping showed that the Cr oxide layer had a corundum crystal structure with C-axis orientation in the direction normal to the substrate. However, a domain was observed at 60 degrees to the in-plane direction. The results showed that a C-axis oriented layer of α -Cr₂O₃ was formed on the substrate surface.

(2) Formation of α -Ga₂O₃ Film

An α -Ga₂O₃ film was formed in the same manner as in Example 1 on the composite base substrate formed in (1), and various evaluations were performed. The ratio R was 0.000. The entire film surface was observed in the same manner as in Example 1, and a large number of cracks were observed. Application of a predetermined voltage to the

Schottky barrier diode produced in the same manner as in Example 1 in the reverse direction caused a leakage current much larger than that in Example 4.

Table 1 summarizes the results for Examples 1 to 5 and Comparative Examples 1 and 2.

5

TABLE 1

	R	Cracks	Leakage current
Example 1	0.053	None	None
Example 2	0.192	None	None
Example 3	0.583	None	Slightly larger than that in Example 1
Example 4	0.692	None	Slightly larger than that in Example 3
Example 5	0.008	None	None
Comparative Examples 1 and 2	0	Large number	Much larger than that in Example 4

10

15

What is claimed is:

1. An α -Ga₂O₃ semiconductor film having a dark spot, which is defined as a measurement point with a maximum emission intensity A of not more than 0.6 times an average value X of top 5% of the maximum emission intensities A at all measurement points in intensity mapping of plane cathodoluminescence, wherein the maximum emission intensity A at each measurement point is determined in a wavelength range of 250 to 365 nm.

20

25

2. The α -Ga₂O₃ semiconductor film according to claim 1, wherein a ratio R of the number of dark spots to the number of all measurement points ranges from 0.005 to 0.200.

3. The α -Ga₂O₃ semiconductor film according to claim 2, wherein the ratio R ranges from 0.050 to 0.200.

30

* * * * *